LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Diagram Checklist: Draw Each One Twice Before the Exam

Diagrams carry easy marks if the **labels** are right. Draw each one from memory, check the labels against your textbook, and tick the box. Use a pencil, keep label lines straight and parallel, and never let them cross.

Biology

Diagram	Ch	Labels you must show	✔	✔
Human digestive system	5	Mouth, oesophagus, stomach, liver, gall bladder, pancreas, small intestine, large intestine, rectum, anus		
Open and closed stomata	5	Guard cells, stomatal pore, epidermal cells, chloroplast		
Human respiratory system	5	Nasal passage, pharynx, larynx, trachea, bronchi, bronchioles, lungs, alveolar sacs, diaphragm, rib		
Section of the human heart	5	Right & left atrium, right & left ventricle, aorta, pulmonary artery, pulmonary veins, vena cava		
Human excretory system	5	Kidneys, renal artery & vein, aorta, vena cava, ureters, urinary bladder, urethra		
Nephron	5	Glomerulus, Bowman's capsule, tubule, branch of renal artery & vein, collecting duct		
Neuron	6	Dendrite, cell body, nucleus, axon, nerve ending		
Reflex arc	6	Receptor, sensory neuron, spinal cord (relay neuron), motor neuron, effector muscle		
Human brain	6	Forebrain (cerebrum), midbrain, hindbrain: cerebellum, pons, medulla; pituitary		
Binary fission (Amoeba) / budding (Hydra)	7	Nucleus dividing, daughter cells / bud, parent		
Longitudinal section of a flower	7	Petal, sepal, stamen (anther, filament), pistil (stigma, style, ovary), ovule		
Pollen tube growing on the stigma	7	Pollen grain, stigma, style, pollen tube, ovary, female germ cell		
Human reproductive systems	7	Male: testis, scrotum, vas deferens, seminal vesicle, prostate, urethra. Female: ovary, oviduct, uterus, cervix, vagina		

Physics

Diagram	Ch	Labels you must show	✔	✔
Concave mirror: all 6 object positions	9	P, F, C, principal axis, two rays with arrows, image		
Convex lens: all 6 object positions	9	O, F_1, F_2, $2F_1$, $2F_2$, two rays with arrows, image		
Convex mirror and concave lens	9	Virtual rays dotted; image behind mirror / same side as object		
Refraction through a glass slab	9	Incident ray, normals, refracted ray, emergent ray, i, r, e, lateral displacement		
Human eye	10	Cornea, iris, pupil, eye lens, ciliary muscles, retina, optic nerve		
Myopia and hypermetropia with correction	10	Image position, correcting lens, far / near point		
Dispersion by a prism	10	White light, prism, spectrum VIBGYOR, red on top, violet at bottom		

Series and parallel circuits	11	Cell, key, resistors, ammeter (in series), voltmeter (in parallel), symbols		
Field around a bar magnet	12	N, S, field lines with arrows going from N to S outside the magnet		
Field around a straight wire / solenoid	12	Current direction, concentric circles / parallel lines inside, N and S ends		
Domestic electric circuit	12	Live, neutral, earth wires, fuse, meter, main switch, 5 A and 15 A lines		

Chemistry

Diagram	**Ch**	**Labels you must show**	✔	✔
Electrolysis of water	1	Carbon electrodes, test tubes with H_2 (double volume) and O_2, battery, water + dilute acid		
Zinc + acid, testing hydrogen	2	Test tube, zinc granules, dilute acid, delivery tube, soap bubbles, burning candle		
Electron-dot structures	3, 4	NaCl, $MgCl_2$ (transfer); H_2, O_2, N_2, CH_4, CO_2 (sharing)		
Electrolytic refining of copper	3	Impure Cu anode, pure Cu cathode, acidified $CuSO_4$, anode mud		
Soap micelle	4	Hydrophilic heads outside, hydrophobic tails inside, oily dirt in the centre		